

AapdaNetra: NDMA Data Integration & Architecture Dossier

Authoritative Datasets, CAP SMS Protocol, Civil Defense SOPs & Resource Mapping

Prepared for: Smart India Hackathon (SIH) Evaluators, NDMA/SDMA Authorities & Incident Commanders
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AapdaNetra (* & > (G \$ M 0) is engineered from the ground up to integrate with and extend frameworks established by the National Disaster Management Authority (NDMA), Government of India. This document details the exact NDMA guidelines, authoritative governmental telemetry pipelines, Common Alerting Protocol (CAP) compliance, and emergency resource directories utilized across the platform.

Key Takeaway: Relationship Between AapdaNetra and NDMA

AapdaNetra does not replace NDMA; rather, it provides the missing predictive artificial intelligence (AI) and operational decision-support layer on top of NDMA frameworks. While existing portals (like NDMA SACHET) broadcast historical weather warnings, AapdaNetra computes sequence-to-sequence (+2h to +24h) temporal flood curves, tracks real-time shelter bed carrying capacities, and navigates citizens around inundated culverts.

1. Statutory NDMA Frameworks & Standards Implemented

CORE COMPLIANCE

- AapdaNetra Risk Categorization Matrix** maps NDMA's 4-color hazard grading framework mapped to a 0–100 threat index: Green (0–20: Normal/Baseline), Amber (21–50: Alert/SOP Tier-1), Red (51–75: Warning/SOP Tier-2), and Critical (76–100: Emergency Evacuation/SOP Tier-3).
- National Hazard Atlas Alignment**: Criticality scoring for drainage siphon contours, and landslide cut-slope hazard perimeters across Delhi, Patna, Rewa/Vindhya, Mumbai, Ranchi, and Guwahati are seeded directly from published NDMA Hazard Atlas surveys.
- National Disaster Management Plan (NDMP 2019/2024)**: Resource Allocation System (IRS) established by NDMA guidelines, matching resource deployment stages to district magistrate control protocols.
- Common Alerting Protocol (CAP) and DoT (1393)**: Push sentinel alerts comply with India's National CAP Standard adopted by NDMA and the Department of Telecommunications (DoT).

2. Authoritative Datasets, Telemetry APIs & Resources

DATA INGESTION

AapdaNetra synthesizes ground-truth data from six primary governmental and scientific sources into a unified geospatial intelligence vector:

Source / Agency	Domain / Parameter	Update Rate	Integration Endpoint / Resource URL
NDMA (Govt of India)	Hazard Atlas Polygons, SOP Guidelines	Static / Baseline	https://ndma.gov.in • National Disaster Management Plan
CWC (Central Water Comm.)	River Gauge telemetry, Danger Stage	15–60 Mins	https://cwc.gov.in • National Hydrology Project (NHP)
IMD (India Met Dept)	Doppler Radar, Precipitation, Nowcasts	15 Mins	https://mausam.imd.gov.in • IMD Open Telemetry API
ISRO Bhuvan / NRSC	CartoDEM, Elevation Gradients, Slope	Static / Geodetic	https://bhuvan.nrsc.gov.in • ISRO Disaster Management Support
Census GIS India	Demographic density, vulnerable clusters	Decennial / Est.	https://censusindia.gov.in • Ward GIS Boundary Maps
OpenStreetMap / Overpass	Road network topology, culvert locations	Continuous Sync	https://www.openstreetmap.org • Overpass QL Hydrography

3. Official Emergency Helplines & Direct Routing

CITIZEN ACCESS

AapdaNetra’s AI Emergency Assistant and Emergency Contacts Modal embed direct, one-tap connection to verified statutory national and state disaster helplines:

Service / Organization	Helpline Number	Operational Scope	Mandate
NDMA National Disaster Control Room	178	Pan-India 24x7	National Disaster Management Authority Helpline
State Disaster Management Authority (SDMA)		State-Level Operations	State Emergency Operations Center (SEOC)
District Disaster Management Authority (DDMA)		District Control Room	Immediate field response coordination
Emergency Response Support System (ERSS)		Unified Emergency	Police, Fire, and Ambulance unified dispatch
Ambulance / Emergency Medical Triage	102 / 102	Medical Emergency	Critical patient and trauma transport

4. Certified NDMA Standard Operating Procedures (SOPs)

AI GROUNDING

AapdaNetra’s Retrieval-Augmented Generation (RAG) assistant is strictly grounded on official civil defense standard operating procedures to eliminate AI hallucinations:

- Flood Management SOP (NDMA Guidelines):** Mandatory food stockpiles, cutoff, elevated non-perishable food storage, and mandatory relocation when gauge levels breach High Flood Level (HFL). Prohibits walking or driving through standing water deeper than 15cm.
- Earthquake Safety SOP:** "Drop, Cover, and Hold On" under rigid furniture, isolate domestic LPG gas lines, avoid elevators, and await certified structural safety clearances before re-entering masonry buildings.
- 72-Hour Survival Kit Standard:** Minimum 3L potable drinking water per person per day, water purification tablets (Halazone/chlorine), waterproof pouches for Aadhaar/property deeds, high-calorie food, battery-powered transistor radios, and basic trauma first-aid dressings.

5. Comparative Analysis: AapdaNetra vs. NDMA SACHET Portal

INNOVATION MATRIX

Capability / Feature	NDMA SACHET Portal	AapdaNetra Platform (Our System)
Alert Nature	Passive broadcast of issued weather alerts	Predictive hydrodynamic simulation (+2h to +24h)
Machine Learning	Rule-based alert relaying from IMD feeds	Sequence-to-Sequence GRU Neural Networks + XGBoost
Shelter Intelligence	Static list of government buildings	Dynamic vacancy tracking with carry-capacity deficit alerts
Evacuation Routing	General safety tips; static directions	Turn-by-turn flood-safe routing bypassing submerged culverts
Citizen Interface	Static English/Hindi alert push notification	Interactive Multilingual AI Assistant (Hindi, Hinglish, Bengali, Assamese)
Explainable AI (XAI)	Not available (Black-box warning)	SHAP factor decomposition explaining WHY risks are rising
"What-If?" Simulation	Not available	Interactive stress-testing sandbox with real-time shelter deficit modeling

Institutional Recommendation & Hackathon Pitch Summary

AapdaNetra is architected as an operational force-multiplier for NDMA and SDMA officials. By ingesting existing NDMA hazard polygons and IMD radar feeds, it automates the complex cognitive workload of incident commanders—calculating exactly how many shelter beds will be short, which culverts will drown, and sending verified cell broadcast SMS to endangered citizens.